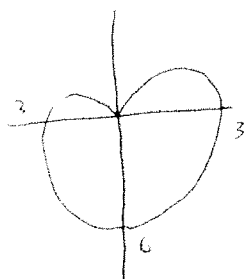


1. Graph the function $r = 3(1 - \sin(\theta))$.

r	θ
3	0
0	$\pi/2$
3	π
6	$3\pi/2$

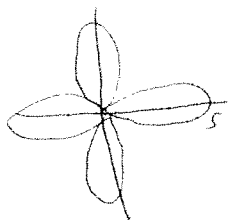


2. Set up an integral to find the area enclosed by $r = 3(1 - \sin(\theta))$.

$$\int_0^{2\pi} \frac{1}{2} (3(1 - \sin(\theta)))^2 d\theta$$

3. Find the area enclosed by three leaves of the four-leafed rose $r = 5 \cos(2\theta)$.

r	θ
5	0
0	$\pi/4$
-5	$\pi/2$
0	$3\pi/4$



$$3 \int_{-\pi/4}^{\pi/4} \frac{1}{2} (5 \cos(2\theta))^2 d\theta \quad \text{[or symmetry]} \quad 3 \int_0^{\pi/4} \frac{1}{2} 25 \cos^2(2\theta) d\theta$$

4. Evaluate $\int \cos^2(3x) dx$.

Recall $\cos^2 \theta = \frac{1}{2} (1 + \cos(2\theta))$. So

$$\int \cos^2(3x) dx = \int \frac{1}{2} (1 + \cos(6x)) dx = \frac{1}{2} \left(x + \frac{1}{6} \sin(6x) \right) + C$$

5. Set up an integral to find the length of the curve $y = 4 \ln x$, from $x = 1$ to $x = 3$

(a) By integrating with respect to x .

$$\int_1^3 \sqrt{1 + \left(\frac{4}{x}\right)^2} dx$$

(b) By integrating with respect to y .

$$y = 4 \ln x \quad \text{so} \quad x = e^{\frac{1}{4}y}$$

$$\int_0^{4 \ln 3} \sqrt{1 + \left(\frac{1}{4} e^{\frac{1}{4}y}\right)^2} dy$$

6. Set up an integral to find the area of the surface generated by rotating the curve $y = 4 \ln x$, from $x = 1$ to $x = 3$

(a) About the x -axis.

$$\int_1^3 2\pi (4 \ln x) \sqrt{1 + \left(\frac{4}{x}\right)^2} dx$$

(b) About the line $y = -3$.

$$\int_1^3 2\pi (4 \ln x - (-3)) \sqrt{1 + \left(\frac{4}{x}\right)^2} dx$$

(c) About the line $x = -3$.

$$\int_0^{4 \ln 3} 2\pi (e^{\frac{1}{4}y} - (-3)) \sqrt{1 + \left(\frac{1}{4} e^{\frac{1}{4}y}\right)^2} dy$$